

## HOMEWORK 3 AND 4

### DUE JUNE 12, 2026

Full credit (30 points) can be obtained by adequately answering 4 of the following questions. Extra credit points are available for exceptional additional effort. Acknowledge any collaboration and LLM use.

#### Final exam review.

**Exercise 0.1.** State the full definition of an (o-minimal) structure on  $\mathbb{R}$ . (All axioms (S1)-(S4), (O1)-(O2), and (S5)-(S6))

**Exercise 0.2.** Explain what a *definable set* and *definable function* in a structure is. Describe some strategies for showing that a set/function is definable.

**Exercise 0.3.** Draw a picture of the universe of structures on  $\mathbb{R}$ . Be sure to include the structures  $\mathbb{R}_{\text{alg}}$ ,  $\mathbb{R}^{\text{RE}}$ ,  $\mathbb{R}_{\text{an}}$ ,  $\mathbb{R}_{\text{exp}}$ ,  $\mathbb{R}_{\text{an,exp}}$ ,  $\text{Pfaff}(\mathbb{R}_{\text{alg}})$ ,  $\text{Pfaff}(\mathbb{R}_{\text{an}})$ ,  $(\mathbb{R}_{\text{alg}}, 2^{\mathbb{Z}})$ ,  $\mathbb{R}_{\text{PH}} = (\mathbb{R}_{\text{alg}}, \mathbb{Z})$ , and  $\mathcal{R}^{\top}$ . In particular:

- define these structures,
- indicate all relevant extensions between these structures,
- indicate the main dividing lines: o-minimal, polynomially bounded, and exponentially bounded,
- indicate the field of exponents of these structures, and
- indicate examples of functions (e.g., activation and loss functions) which either are or are not definable in each of these structures.

**Exercise 0.4.** State Rademacher's Theorem for locally lipschitz functions and the definition of the Clarke subgradient for a locally lipschitz function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ . Prove or sketch an argument why the Clarke subgradient is definable in any structure satisfying axioms (S1)-(S4) and (S5)-(S6), provided  $f$  is definable in that structure.

**Exercise 0.5.** Generate 5 True/False questions on the material from the lecture and discussion; then answer them.

#### Other exercises.

**Exercise 0.6.** Draw some pictures/diagrams illustrating some of the ideas from the course so far.

**Exercise 0.7.** Suppose  $X \subseteq \mathbb{R}^2$  is definable in  $\mathbb{R}_{\text{an,exp}}$ . Suppose  $X \cap \Gamma(\Gamma)$  is infinite, where  $\Gamma(\Gamma)$  is the graph of Gamma function  $\Gamma : (0, +\infty) \rightarrow \mathbb{R}$ . Prove that  $X$  has nonempty interior.

**Exercise 0.8.** Suppose  $\mathcal{R}$  is a structure on  $\mathbb{R}$  and  $V \subseteq \mathbb{R}^n$  is an arbitrary  $\mathbb{R}$ -subspace.

- (1) Show that  $V$  is definable.
- (2) Suppose  $d$  is a dimension function on  $\mathcal{R}$ . Show that:

$$\dim_{\mathbb{R}} V = d(V)$$

where  $\dim_{\mathbb{R}}$  denotes the usual  $\mathbb{R}$ -linear dimension of  $V$  from linear algebra.

**Exercise 0.9.** Suppose  $\mathcal{R}$  is an o-minimal structure on  $\mathbb{R}$  and  $M \subseteq \mathbb{R}^n$  is a  $C^1$ -manifold of dimension  $d$  (i.e.,  $d = \dim_{\mathbb{R}} T_M(x)$  for every  $x \in M$ ). Further suppose  $M$  is definable as a subset of  $\mathbb{R}^n$  and  $d$  is a dimension function on  $\mathcal{R}$ . Show that:

$$d = d(M)$$

Is this still true without the o-minimal assumption?

**Exercise 0.10.** Suppose  $\mathcal{R}$  is an o-minimal structure on  $\mathbb{R}$  and  $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is a definable function. Given a nonempty definable open set  $C \subseteq \mathbb{R}^n$ , show that there exists a nonempty definable open set  $C' \subseteq C$  such that either  $F|_{C'} : C' \rightarrow \mathbb{R}^n$  is a homeomorphism onto its image, or  $\dim F[C'] < n$ .

## REFERENCES